LEARNING OUTCOMES

After learning this Chapter, the student will

1. **Apply the concept of special type semiconductor devices to design various circuits.**
2. **Acquire the basic knowledge in opto-electronic devices and Thyristor devices.**
3. **Understand how to control the power using Thyristor type devices.**
4. **Design, fabricate and test small electronic circuit.**

GLOSSARY

S. No	Terms	Explanation
1	Diffusion	The intermingling of substances by the natural movement of their particles
2	Electroluminescence	Luminescence produced electrically by the application of a voltage
3	Gate	The controlling terminal of a FET. A voltage on the gate control the current flow between the source and drain
4	Indicators	A thing that indicates the state or level of something
5	Intensity	The measurable amount of a property such as force, brightness or a magnetic field
6	Monolithic	Solid state circuit composed of active and passive components formed in a single chip
7	Nematic	Relating to or denoting a state of liquid crystal in which the molecules are oriented in parallel but not arranged in well-defined planes
8	Optoelectronics	Technology concerned with the combined use of electronics and light
9	Q Factor	A measure of the quality of a resonant circuit
10	Radiation	The emission of energy as electromagnetic waves or as moving subatomic particles especially high energy particles which causes ionization
11	Selenium	It is a grey crystalline non-metal with semiconductor properties
12	Substrate	An underlying substance or layer
13	Terminal	A point of connection for closing an electric circuit
14	Threshold	The magnitude or intensity that must be exceeded for a certain reaction to occur
15	Thyristor	A four layered semiconductor rectifier in which the flow of current between two electrodes is triggered by a signal at a third electrode

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